

Review

Talent Identification and AI-Driven Decision Tools in Sport: A Policy-Oriented Perspective on Algorithmic Bias, Data Privacy, and Digital Determinism in Player Evaluation

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Abstract

Big-data analytics are increasingly used in scouting and talent identification, with machine learning (ML) tools applied to evaluate and predict player performance based on match statistics, video tracking, physical and anthropometric tests, psychological assessments, social media data, and qualitative scouting reports. Advances in computer vision, together with the emergence of affordable automated broadcasting and data collection systems, have extended the deployment of ML-driven scouting from professional to youth sport. The use of algorithms in educational, employment, and healthcare settings has been shown to introduce biases and discrimination while wrongly assuming accuracy and objectivity because the decisions are made automatically and quantitatively. In this respect, we briefly describe the development of data-driven performance analysis and how ML-based technologies are currently applied for early screening and comparison of large player populations. Based on a narrative overview of the literature, we draw on evidence from education, employment, and healthcare to identify risks that may also emerge in ML-driven player evaluation, including algorithmic bias, non-representative training data, privacy concerns, and the persistence of model-based labels over time, especially in youth sport. Our main contribution is translating these threats into governance principles and operational safeguards for responsible use of AI in scouting and talent identification.

Keywords: AI-driven decision tools; algorithmic bias; data privacy; digital determinism; scouting; sport analytics; talent identification



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1. Introduction

Sports analytics can be defined as the systematic collection, organization, analysis, dissemination, and interpretation of data to evaluate performance, inform tactical and strategic decisions, and support processes such as scouting, player development, and resource allocation, with the objective of achieving competitive advantage [1]. Similar to the scientific method, sport analytics often progresses through four linked stages: description, explanation, prediction, and control. It begins with the systematic collection and analysis of data on events occurring on and off the field, using methods ranging from basic summaries (e.g., means and proportions) to unsupervised learning techniques and data visualization [2]. As data accumulate, the focus shifts toward explanation by identifying recurring patterns and statistical relationships between variables, such as links between technical

ability tests and individual or team performance indicators [3]. Once robust associations are established, these insights can be extended to prediction of future outcomes such as injuries or performance [4]. The final stage is prescription, in which analytical findings are used to deliberately influence performance, for example by adjusting training loads or introducing tactical changes to exploit opponent weaknesses [5].

Aligned with the broader domain of business analytics, sport analytics inform decision-making across multiple functional areas within the sport industry, including fan engagement, ticket pricing, demand forecasting, and merchandise sales [6]. Within this broader framework, a performance-oriented, on-field subdomain—commonly termed *performance analysis*—focuses specifically on the use of player and team data to enhance outcomes on the field of play. Its core function is to identify meaningful relationships among variables and to translate these relationships into actionable insights regarding individual and collective performance [7,8].

1.1. The Emergence of Tracking Systems and Big-Data-Based Performance Analysis

The National Basketball Association (NBA) partnered with IBM, which deployed trained personnel equipped with dedicated computers to code game events and construct a centralized database. By the 1995–1996 season, IBM introduced the *Advanced Scout* software, which enabled structured data analysis through a user-friendly interface that was gradually adopted across the league [9]. In football, automated tracking platforms emerged in the mid-1990s with the development of the *Amisco* system. The system captured player, ball, and referee positions multiple times per second using object-recognition algorithms, while human operators simultaneously coded game actions (i.e., tagging) such as passes, shots, and tackles. This hybrid approach produced detailed event-level datasets that enabled the analysis of tactical, technical, and workload aspects. This combination of automated tracking accompanied by human tagging of events remains standard across invasion sports to this day. Another prominent example is the *SportVU* tracking system, deployed across NBA arenas and used league-wide from the 2013–2014 season through *STATS* platform [10].

Major League Baseball (MLB) also employs an advanced tracking infrastructure called *Statcast*. Introduced in 2015 using a combination of Doppler radar and high-speed cameras and later upgraded to a fully optical multi-camera system, this platform captures high-frequency spatiotemporal data on both the ball and players, including pitch velocity and spin rate, batted-ball characteristics such as exit velocity and launch angle, and player positioning, sprint speed, and defensive range, which all support player evaluation, tactical preparation, and load management [11]. These technological developments signify a transition, from around 2010 onward, toward machine learning (ML) application within large-scale data collection systems in professional sport such as *PlaySight* in tennis, *Opta* in rugby, *PlayVision* in volleyball, and *Sportradar* in handball.

Nowadays performance analysts are a mainstay of any coaching staff in professional sport [12]. Analysts rely on sophisticated tools such as semi-automated tracking systems and ML-based computer vision to quantify training and competition characteristics in a valid and reliable manner [13,14]. Additionally, the global positioning system (GPS) device, pedometer, accelerometer, and gyroscope wearable sensors allow the collection of diverse technical and load information. These data are being supplemented by physiological sensors (heart rate, heart rate variability, skin temperature, skin conductance response, sleep, etc.) and subjective wellness questionnaires. ML-based tools are being deployed to aggregate assessments of physical characteristics such as speed, endurance, movement intensity, and recovery rates with technical indicators including passing accuracy, ball possession, dribbling, and shooting efficiency. The outputs of these models support applications such

as overall load monitoring, identification of elevated injury risk, and informed adjustment of nutrition and rehabilitation strategies [15,16].

On the tactical level, analysts use advanced statistical procedures to determine factors leading to a win in a match and to identify *key performance indicators (KPIs)* that distinguish between successful and less-successful players, teams, and, in some cases, leagues as a whole [17]. Tactical factors such as possession, ball movement, passes completed, and players' usage of space are evaluated to detect patterns leading to scoring events [18]. Modern match analysis therefore consists of the calculation of detailed metrics describing players' interactions in space and time, including team spread and coverage area, synchronization processes, predictability of player movement, and the entropy of passing patterns [19].

Importantly, in the sport science community the term *artificial intelligence (AI)* is commonly used as an umbrella label for advanced data-driven technologies or big-data-based analytics, frequently without a precise definition. Hammes et al. [20] describe an intelligent machine as one that interacts with its environment by perceiving it, constructing a model, generating a plan, and executing an action. Within this broader framework, *machine learning (ML)* refers to algorithms that detect statistical regularities in data and use them to make predictions or classifications without relying on predefined rules. Rather than being explicitly programmed, model parameters are estimated by optimizing performance on observed data, enabling generalization to new cases. This approach allows continuous refinement as additional data become available, improving predictive accuracy and stability over time. In practice, most tools labeled as AI in sport rely primarily on ML methods, as the scale, complexity, and variability of movement and action data favor statistical learning over rule-based or symbolic systems. For this reason, we use the term ML throughout this paper when referring to data-driven performance analysis tools, including player evaluation, while reserving the term AI for broader big-data-based decision-support systems.

1.2. Player Evaluation and Scouting

For much of the twentieth century, professional clubs relied on networks of scouts whose work consisted of repeated live observation, note-taking, and the production of standardized and non-standardized reports on potential recruits. These evaluations focused on technical skills, physical attributes, tactical understanding, and perceived psychological qualities, and were used to compare candidates and to identify players who met positional needs and budget constraints [21]. Attributes were typically assessed using structured rating scales and checklists; however, final prediction often relied on intuitive integration of multiple performance indicators rather than standardized combined rankings [22].

During the 1980s, *Bill James* drew on large volumes of box-score data to question conventional scouting statistics such as batting average and runs batted in. He proposed alternative metrics intended to better reflect player contribution and developed early predictive models that applied age adjustments, historical comparisons, and regression-based reasoning to generate forward-looking performance predictions [23]. James's insights were published independently in his *Baseball Abstracts* and were not readily integrated into professional decision-making, as scouting and roster-construction processes continued to rely primarily on qualitative scouting reports and subjective managerial evaluations [24]. Similarly, in European football, player evaluation and scouting initially relied on qualitative observation, with notational methods gradually evolving to record technical and tactical actions such as passes, shots, tackles, and ball recoveries, allowing performances to be compared across matches [25]. Such statistical records became increasingly incorporated for retrospective performance review and coaching feedback, while also providing a more organized basis for comparing players than subjective evaluation alone [26].

Building on large-scale datasets that had accumulated in professional baseball by the late 1990s, *Nate Silver* identified variables associated with future player performance and developed a predictive model. His *Player Empirical Comparison and Optimization Test Algorithm (PECOTA)* compared each player against a database of roughly 20,000 MLB player-seasons since World War II and produced probabilistic forecasts while accounting for age-related development and regression toward the mean [27]. Silver's approach is often cited as a key step in the evolution of predictive modeling and a landmark of big-data-based analytics in sport, as its forecasts were shown to perform at least as well as, and in some cases better than, traditional expert scouts [23]. PECOTA relied on statistical estimation using historical data. The form of the model, including how similarity was defined, how age effects were applied, and how variables were combined, were specified in advance. In contrast, modern ML approaches estimate variable importance, interactions, and non-linear relationships jointly during training.

Lu et al. [28] extended this shift by using wearable sensors attached to football players' ankles, with the aim of capturing movement patterns during training sessions. Using ML, the system learned to infer players' level directly from these patterns while also distinguishing between different actions, enabling an objective grading of performance. Notably, the system separated elite and amateur players with accuracy above 90%. Another important difference is that traditional methods such as field testing, observational assessment, and expert judgment rely on time-consuming and inherently subjective procedures. In contrast, ML allows us to identify patterns across multiple athlete-performance characteristics through the processing of complex and often nonlinear relationships between technical, physical, physiological, and cognitive factors. In this vein, ML-based approaches offer the advantages of speed, scalability, and greater objectivity, qualities that are particularly relevant in sports contexts where decisions often need to be made under time constraints and with limited opportunities for direct testing [15].

Building on earlier projection models such as PECOTA, recent ML approaches use *artificial neural networks* to analyze player data, enabling the prediction of future league status with relatively high accuracy and providing a more objective basis for recruitment decisions [29]. Accordingly, contemporary scouting relies on retrospective analysis, examining highly successful professional athletes and tracing their developmental histories in order to identify characteristics that distinguish those who reached elite levels from prospects who remained at a sub-elite level [30]. This backward-looking logic is used to assess whether indicators observed at younger ages, such as high school or college, carry meaningful information about later professional performance. Explicit physical performance measures including medicine ball throw, bench press strength, vertical jump height, and sprint speed have been examined as potential predictors of future success, often with attention to positional demands and sport-specific requirements, and are consistently used to gauge athletic potential at the high school and collegiate levels [31,32].

In NBA draft as an example, teams select eligible players according to a predefined order. Early draft positions represent high-impact decisions as analytical units seek to estimate which candidates are most likely to improve team performance over subsequent seasons. To support this task, performance information drawn from play-by-play records, tracking systems, physical testing, anthropometric measures, psychological assessments, and structured scouting reports is organized and visualized to allow comparison across players and competitive contexts. This process is informed by professional staff (i.e., expert judgment) to account for game style, team dynamics, and various coaching priorities through the adjustment of technical actions for playing time and team possession, conditioning physical outputs on match intensity, and normalizing indicators by age, position, and specific squads. ML techniques, informed by this expert input, are subsequently

applied to integrate, weight, and compare multiple data sources simultaneously across players and competitive contexts [33].

In football, global scouting platforms such as *Wyscout*, *InStat*, *Hudl*, and *Transfermarkt* have become central tools for initial identification of potential prospects. These platforms aggregate match video, event data, and contextual information across leagues and age groups, allowing us to monitor large player populations remotely. Analysts apply pre-defined criteria related to age, position, competition level, and performance indicators to reduce large candidate pools to smaller subsets suitable for closer inspection. Event-based measures are commonly adjusted for playing time, team possession, tactical role, and league strength before being used as inclusion criteria. Beyond conventional technical and physical performance metrics, these platforms facilitate the integration of more intricate psychological and physiological characteristics, including *mentality composure*, *movement reaction*, and *power strength* [34].

Leveraging these advancements, Rinaldi et al. [35] analyzed a vast array of player data, including performance statistics, physical attributes, and historical injury records, to identify multidimensional patterns and interactions that are difficult to model using traditional regression-based and manually specified analytical approaches. Their system incorporated similarity techniques to compare prospective recruits with current players, providing a more nuanced understanding of how a new player might fit into existing squad. Mănescu and Mănescu [15] showed that ML tools can meaningfully differentiate between performance levels and identify key predictors in athlete evaluation. Accordingly, outputs from ML-driven models are increasingly used for player ranking, similarity profiling, and performance projection; however, at this stage, they primarily serve initial identification, guiding resource allocation and prioritizing further in-depth observation rather than completely replacing expert judgment. As emphasized across applied studies, in-person evaluation and contextual knowledge remain decisive at advanced stages of recruitment [36].

Despite the rapid integration of ML into performance evaluation, both research and practice have focused more on predictive performance and technical applications than on how such systems should be governed in sport-specific contexts. In particular, limited attention has been given to algorithmic bias, privacy, accountability, and persistent labeling, especially in youth sport where early decisions may shape long-term opportunities [37,38]. Drawing on a narrative review of the literature, rather than a formal systematic or scoping review aimed at exhaustive and reproducible evidence coverage, this policy-oriented paper addresses that gap by examining emerging risks and proposing governance principles for responsible AI use in sport.

2. The Threat of Algorithmic Bias

In occupational settings, ML-based screening systems are increasingly used to improve efficiency and standardization in candidate evaluation. These tools are typically trained on historical hiring and promotion data, which reflect past organizational practices and decision criteria. As a result, models may reproduce existing inequalities when prior success is correlated with gender, race, socioeconomic background, or access to elite institutions, even when protected attributes such as race, gender identity, or religion are excluded from model inputs, because proxy variables in the data can still encode social background, including educational pathways, geographic location, and employment (unemployment) histories [39,40].

Similar mechanisms have been documented in educational assessment and admissions. Predictive models used to estimate academic success or dropout risk are typically trained on standardized test scores, attendance records, and prior grades, which often reflect

accumulated educational advantage rather than learning potential. As a result, students with interrupted schooling, delayed development, or non-linear learning trajectories are more likely to be classified as high risk, limiting access to scholarships and selective programs. In systems where early performance metrics shape later opportunities, such models can reinforce existing disparities over time [41].

In medical contexts, risk-prediction algorithms trained on historical treatment and utilization data have produced comparable distortions. A widely cited example is the study by Obermeyer et al. [42], which demonstrated that a healthcare risk-prediction model trained on historical expenditure data systematically underestimated clinical need among disadvantaged populations. Because the algorithm used prior healthcare spending as a proxy for illness severity, and historically lower spending reflected unequal access to care rather than lower disease burden, the model learned to predict cost instead of need, thereby assigning lower risk scores to sicker patients from disadvantaged groups.

Bullock et al. [4] warn that ML methods tend to be opaque, as their underlying architectures are typically too complex to fully disentangle predictor–outcome relationships and the availability of implemented software to obtain individualized predictions is rarely seen. Such predictive tools are often referred to as “black boxes,” as they are not easily interpretable and the reasons underlying their outputs are not explicitly clear. Importantly, due to the commercial nature of many predictive tools, developers are often reluctant to transparently report or openly share the development and validation of their algorithms [43].

Facing these concerns, the *Federal Trade Commission (FTC)* warned already in 2016, in its report (<https://www.ftc.gov/system/files/documents/reports/big-data-tool-inclusion-or-exclusion-understanding-issues/160106big-data-rpt.pdf> accessed on 8 March 2026) “*Big Data: A Tool for Inclusion or Exclusion, Understanding the Issues*,” about the legal risks associated with ML analytics. The report highlighted the potential for bias and discrimination at every stage of the analytical process, from dataset selection and problem formulation to decision-making based on model outputs. These issues are particularly salient in recruitment and selection tools [40]. Since then, concerns about algorithmic fairness and accountability have informed broader federal policy efforts, including initiatives under the *National AI Initiative Act* and the establishment of the *National Artificial Intelligence Research Resource Task Force*, which addresses issues related to security, privacy, access, and civil rights in the development and governance of AI systems (<https://www.nsf.gov/news/biden-administration-launches-national-artificial#:~:text=As%20directed%20by%20Congress%20in,all%20scientific%20disciplines%20with%20access> accessed on 8 March 2026).

It is reasonable to assume that risks observed in occupational and educational settings will also be present in talent identification in sport, where ML models similarly rely on historical success labels that often reflect access, opportunity, and prior selection rather than underlying potential. As a result, dominant athlete profiles embedded in historical data lead models to translate past selection patterns into automated decision rules, narrowing the range of prospects considered and overlooking alternative development routes [21,44].

A simple example can be seen in the social backgrounds associated with different sports. In many European countries and North America, boxing has traditionally been linked to lower socioeconomic and working-class environments [45]. Models trained on data from elite boxers may therefore learn characteristics of these backgrounds (e.g., family size, neighborhoods, schools) as indicators of success. If interviews, social media, or behavioral characteristics are included in training data, language and communication styles common in these groups may also become embedded in the model. In contrast, golf and tennis have historically been associated with well-off and predominantly White

populations. Models trained on elite golfers and tennis players may therefore identify features linked to this social background as meaningful predictors. As a result, race- and class-related features can enter the model through proxy variables, even when direct racial and demographic attributes are excluded.

The described dynamics can lead to systematic exclusion of athletes whose developmental pathways diverge from dominant norms, such as late bloomers, players from lower-tier competitions, or those without early access to high-level clubs. As mentioned earlier, the fairness of ML-based decision tools depends heavily on the quality and representativeness of the training data. In sport, this challenge is intensified by a persistent gender bias in historical datasets, often described as the “data gap” [46]. Systems trained predominantly on data from elite male sport frequently generalize poorly to other populations, including youth athletes, female players, and Paralympic competitors. The structural overrepresentation of elite male athletes, combined with systematic omissions such as growth-related developmental changes in youth (e.g., on relative age effect in both group and individual sports see [47,48]) or menstrual cycle effects in female athletes, can distort predictive models and lead to unfair outcomes [49].

Similarly, in sport medicine and load monitoring, predictive models are often trained on datasets that may reflect existing social, physiological, and gender-based disparities in training exposure, injury reporting, and access to care. This may lead to unequal treatment, for example by normalizing higher workloads in certain groups or underestimating injury risk in others. A vivid example would be a model trained on load and wellness data from domestic players in the English Premier League who reliably report ratings of perceived exertion and engage openly with conditioning staff. When applied to foreign players who tend to underreport fatigue due to cultural norms, perceived job insecurity, or communication barriers, such models are prone to underestimate true load and recovery deficits. Over-reliance on model outputs may also reduce clinical judgment and individualized decision-making. In the absence of clear governance procedures and accountability, performance optimization goals may take precedence over considerations of informed consent, data privacy, and athlete agency [50].

3. Data Privacy Concerns

While much attention has focused on algorithmic biases, equally important concerns relate to data privacy and governance, reflecting structural power asymmetries between athletes and the organizations that collect, process, and control performance and health data [37]. In high-performance settings, institutional stakeholders such as teams, leagues, clubs, and technology vendors typically own or manage the resulting datasets, while athletes have limited influence over their use. This imbalance is particularly evident when contractual obligations require compliance with tracking technologies or biometric monitoring as a condition of participation [51].

The use of data-driven selection tools also raises privacy concerns when personal information is collected from athletes’ digital environments. Athletes across age groups routinely share information online through professional profiles, social media platforms, comments, and other forms of digital engagement that they would not voluntarily disclose to recruiters and that organizations would not typically be permitted to request directly. Automated social media screening or *social profiling* provides organizations with rapid and scalable access to unfiltered online content, enabling systematic evaluation of behavioral and personality indicators that were previously difficult to observe [52]. The large-scale aggregation and analysis of such data blur the boundaries between public and private domains introducing new risks related to consent, transparency, and secondary use of personal information. Moreover, such systems may reinforce digital inequality

by disadvantaging individuals who are less active online or who differ in social media literacy and digital self-presentation, thereby introducing new forms of bias and exclusion in selection processes [53].

Outside the domain of sport, the *HireVue* recruitment platform, as an example, offered ML-driven video interview assessments. The system analyzed large volumes of data points related to candidates' voice, word choice, and facial movements to predict traits such as willingness to learn and personal stability. Concerns were raised that these assessments were biased, unprovable, and not replicable, potentially leading to discriminatory hiring decisions. Regulatory intervention was sought to halt automatic scoring and require disclosure of the underlying algorithms and evaluation criteria. In early 2021, HireVue revised its use of facial recognition analysis but continued analyzing biometric data, including speech patterns, intonation, and behavioral indicators (<https://epic.org/hirevue-facing-ftc-complaint-from-epic-halts-use-of-facial-recognition/> accessed on 8 March 2026).

Growing concerns over such tools have prompted regulatory responses through the *General Data Protection Regulation (GDPR)* and the *EU Artificial Intelligence Act*, which impose requirements related to lawful data processing, purpose limitation, data minimization, transparency, and safeguards. The EU classifies certain AI applications as high-risk, subjecting them to mandatory compliance requirements, particularly when they may affect individuals' health, safety, or fundamental rights. Within this framework, privacy and personal data protection require particular attention, as the use of biological and performance data in sport for applications such as doping detection, training optimization, and injury prevention poses a substantial threat.

While AI-based technologies offer clear benefits in enhancing performance and safeguarding athlete health, implementing ethical safeguards that adequately protect fundamental rights remains challenging, especially given the sensitive nature of biological data in sport contexts [38,54]. In this respect, professional sport has largely operated in a regulatory grey zone, where extensive monitoring of athletes' physical, behavioral, and digital data is widely viewed as necessary given the extremely high stakes of elite-level competition [55]. Reflecting this gap, recent EU policy work on AI in sport highlights both the benefits and risks of these technologies and offers initial recommendations for their responsible use (<https://op.europa.eu/en/web/general-publications/ai-2025> accessed on 8 March 2026).

4. Early Talent Identification and Digital Determinism

Identifying young athletes with the potential to excel at the highest level remains an important challenge for coaches, clubs, and national federations [56]. It also remains an open question in sport literature as precise criteria for talent identification (e.g., ball control, passing skills, game awareness, sprinting, strength, explosiveness, confidence, toughness, etc.) are remarkably difficult to isolate [57]. Early performance often reflects temporary advantages related to maturation and access to resources rather than long-term potential. Nevertheless, professional staff are expected to identify promising individuals at increasingly younger ages, despite the limited stability and predictive value of early indicators [58].

In football academies, as an example, GPS-based data gathering systems are now deployed with players as young as 10 [59]. These systems provide continuous measures of movement patterns and overall workload, complementing traditional assessments such as field-based aerobic capacity tests including the Yo-Yo or multistage shuttle run, 30 m sprint and vertical jump tests for athletic ability, structured technical skill evaluation (e.g., passing accuracy, ball control, slalom dribbling), and anthropometric profiling of height, body mass, and expected growth [22].

Yet, the non-linear nature of developmental trajectories in children, together with differences in prior exposure to skill-enriching environments, makes it difficult to disentangle underlying potential from observed performance when relying solely on quantitative data. Maturation-age differences (e.g., late bloomers) and informal play opportunities such as growing up as a younger sibling in a sport-oriented household with easy access to playgrounds complicate benchmarking [60]. For this reason, it has been suggested that talent identification and selection decisions should balance objective metrics with expert subjective evaluation. This balance is particularly important given that 12–13-year-olds already encounter substantial evaluation and selection pressure [61]. Within this context, recent work highlights the need to consider performance across multiple developmental timelines (e.g., relative age, biological age, training age) as each shapes observed outcomes and may introduce bias if overlooked. This supports a call toward a more holistic talent identification approach that combines these factors with broader life pathways, allowing for a better understanding of development over time and a more accurate estimation of long-term potential, rather than relying on single cross-sectional measures [56].

Until recently, broadcasting and large-scale data collection were limited to professional high-budget sport. However, under early pressure for talent identification, recent technological developments now allow the extraction of ball and player spatiotemporal data directly from broadcast video which drastically reduced the costs of data gathering as compared to semi-automated optical tracking systems that required installation of multiple high-speed cameras and experienced human operator who accompanied the system (for a review of ML techniques that allow automatic individual and collective action recognition see [62]). The combination of low-cost broadcast systems (e.g., *Pixellot*, *PlaySight*, *VEO*) with computer-vision data collection platforms brought ML-based analytics to youth sport (<https://www.sportsbusinessjournal.com/Articles/2025/11/05/pixellot-playon-extend-agreement-for-richer-high-school-streaming-solution/> accessed on 8 March 2026). Armed with anthropometric, physiological, performance, and sociodemographic information, algorithms used to optimize talent selection decisions may, if applied uncritically, begin to shape or constrain an individual's sporting trajectory already at middle school level, a dynamic that has been described as digital determinism [55].

Another growing trend in talent identification is ML-based systems that apply computer vision and deep learning to assess athletic and technical ability directly from smartphone footage in youth and amateur sport environments. These tools (e.g., *aiScout*, *Predicta Football*) require athletes to complete standardized drills in everyday settings, enabling remote assessment and comparison against reference performance models (<https://youthsportsbusinessreport.com/ai-ios-ai-talent-discovery-app-reaches-45000-youth-athletes-through-mls-partnership/> accessed on 8 March 2026). By expanding the reach of scouting beyond elite-level academies and national team trials, such platforms aim to improve access, increase efficiency, provide guidance to parents, and promote more objective early talent identification. A variety of such ML-based tools are already being deployed to screen large populations of children in countries such as Senegal, Israel, and Japan (<https://www.bbc.com/news/articles/cmj2jkppvx3o>, <https://aibusiness.com/computer-vision/intel-s-ai-app-scouts-olympic-talent-via-smartphone-camera> accessed on 8 March 2026).

Important notion here is the persistence of algorithmic labels within institutional systems. Once an AI-generated potential profile is recorded, it may continue to shape decisions related to team selection, contract negotiations, or insurance coverage. Rush and Osborne [63] argue that such labels can become self-fulfilling because athletes ranked lower by the model tend to receive fewer opportunities, thereby confirming the original prediction. Commercial pressures further complicate this landscape, as technology

providers developing wearables, data platforms, and proprietary algorithms often retain control over athlete data with limited oversight. In the absence of robust governance, the commodification of performance and health scores may shift the balance from performance support toward data exploitation [50]. Mining youth performance data to build ML-based predictive models is especially concerning at a time when notions of digital dystopia and algorithmic control over our lives are on the rise [64,65].

These risks are closely linked to the regulatory logic of the *General Data Protection Regulation* (GDPR) in the European Union and to specific legal instruments in the United States, such as the *Children's Online Privacy Protection Act* and emerging state biometric privacy laws. The large-scale collection of physiological, biometric, and behavioral data through wearables, computer vision, and digital profiling in youth sport is supposed to be treated as sensitive personal data, requiring explicit parental consent, clear purpose limitation, and demonstrable proportionality [66]. The regulation restricts reliance on fully automated decision-making, which is directly relevant when AI-generated potential scores influence selection, deselection, and access to developmental resources. This obliges organizations to maintain meaningful human oversight, transparent criteria, and the possibility to challenge or reinterpret algorithmic outputs.

A central operational implication concerns data minimization and retention: only data strictly necessary for current developmental support should be collected, and predefined retention schedules must include systematic deletion, anonymization, or aggregation once the original purpose has been fulfilled. This is particularly important given that large youth datasets may enable persistent profiling and predictive modelling beyond the awareness of athletes and their guardians. In practice, the literature highlights a growing gap between formal regulation and its actual implementation, including inadequate infrastructure, limited training of data protection officers, and insufficient awareness among parents and young users regarding their rights and digital risks. Together, these factors increase concerns related to long-term profiling, cross-border data transfer, commercial exploitation of minors' data, and the persistence of algorithmic labels within institutional systems, reinforcing the need for accountable governance and safeguards against exclusion or discrimination in ML-driven early talent identification [67,68].

On top of the aforementioned ethical considerations, from a strictly professional standpoint, the dynamic and continuously co-adaptive nature of team invasion sports further constrains the stability and generalizability of ML-based predictive scores. As tactical patterns, player abilities, roles, and interaction structures evolve in response to opponents and contextual pressures, historical data rapidly lose predictive validity, requiring continuous model recalibration and periodic retraining to remain relevant [69].

Moreover, the opacity of many ML "black box" models, often driven by commercial proprietary considerations, prevents independent evaluation of their validation procedures and limits the ability to determine how scores have been established and whether they generalize beyond the original sample. This is particularly consequential given that predictive models often exhibit inflated performance on training data but fail to maintain accuracy when applied to new or future observations, a pattern commonly associated with overfitting [70]. In such cases, models may be overly adapted to specific patterns in the training data, capturing noise rather than stable signals and thereby reducing their applicability in dynamic sport environments [4].

5. Guidelines for Responsible AI in Player Evaluation and Talent Identification

The technological developments discussed above carry direct implications for core sport values such as inclusiveness and equal opportunity, which are embedded in sport

governance frameworks and legislation [71,72]. At the same time, the absence of clearly defined ethical principles for AI in sport may reinforce existing governance inadequacies, particularly given the relative autonomy of sport governing bodies, which often limits external oversight. In this context, the expansion of AI technologies risks further exposing athletes to asymmetries of power held by teams, federations, and commercial actors. We therefore propose four principles to guide the design and deployment of data-driven systems in sport, aligning algorithmic practices with stakeholder needs while remaining consistent with human rights protection and risk-mitigation requirements. The principles are presented below according to their immediate feasibility and the extent of organizational change required.

5.1. Athlete Data Rights and Privacy-Preserving Design

Athletes, and in youth contexts their legal guardians, should have enforceable rights over biometric, physiological, behavioral, and digital data, including access, correction, portability, and the ability to withdraw consent without punitive consequences. These rights must be embedded within regulatory frameworks that limit data use and any associated predictive outputs to purposes directly related to selection, development, and performance support, while prohibiting secondary uses, such as insurance, employment screening, or commercial profiling, without separate informed consent. The proposed principles should be operationalized through privacy-preserving design, whereby identifiable data remain within the originating organization and only aggregated model updates or statistical summaries are shared across entities. For secondary analyses, strong anonymization must be applied, and access restricted by professional role. Data retention should be predefined and purpose-bound, shorter for youth, with deletion, anonymization, or aggregation once the purpose is fulfilled. Raw youth data should not be stored in open-ended commercial databases, and data processors must be contractually restricted, auditable, and technically prevented from reuse or transfer beyond the stated purpose.

5.2. Decision Support, Transparency, and Human Accountability

ML-based predictive scores should be formally restricted to screening, prioritization, and hypothesis generation. Final selection and deselection decisions should require documented human review, with documentation of how model outputs were considered alongside contextual information and expert judgment. Athletes and, in youth contexts, their guardians should be able to request at least basic information on the main factors shaping the assessment, the degree of uncertainty, and the conditions under which the evaluation might change. Formal procedures should also allow individuals to review, correct, and request reassessment of data or model-based decisions.

5.3. Model Development, Fairness, and Ecological Validation

Model development and ecological validation should be treated as a unified process that addresses both statistical performance and real-world applicability. Validation should extend beyond aggregate accuracy to cover the full modelling pipeline, including data collection, feature construction, training, and testing, with explicit reporting of discrimination, calibration, and subgroup error patterns (e.g., sex, relative age, maturation, and socioeconomic background). When disparities are detected, mitigation strategies such as reweighting, stratified sampling, threshold adjustment, or post hoc recalibration should be applied and transparently reported. Training datasets should be intentionally constructed to reflect diverse populations and developmental contexts, with preference given to longitudinal data that capture trajectories rather than transient states. Internal validation should rely on resampling approaches (e.g., cross-validation or bootstrapping) to obtain unbiased performance estimates and reduce the risk of overfitting.

Different ways of assessing fairness (e.g., fairness metrics) were introduced to examine the distribution of model outcomes across groups such as age cohorts and socioeconomic backgrounds. These metrics evaluate imbalance in error rates across these groups and the consistency of decisions for individuals with similar characteristics. In practice, this involves comparing model outputs and error patterns across relevant subpopulations, identifying systematic discrepancies in treatment, and assessing whether these differences can be justified by task-relevant information (for review on fairness metrics see [73]).

From an ecological validity perspective, model performance should be evaluated across multiple contexts, including different clubs, regions, competition levels, and seasons. Evaluation protocols should prevent any overlap between training and testing samples at the athlete level and explicitly address outcome imbalance, given the rarity of successful career progressions. External validation in the intended population should be treated as a prerequisite for deployment. Finally, models should be recalibrated on a regular schedule to account for evolving tactical patterns, physical demands, and player characteristics, ensuring continued alignment with the dynamic nature of team invasion sports.

5.4. Governance, Oversight, and Regulatory Frameworks

International sport governing bodies should draw on existing regulatory models, such as the World Anti-Doping Agency (WADA), to establish a comparable international structure responsible for oversight of artificial intelligence and athlete data use. Such an agency should include interdisciplinary expertise and have the authority to review, audit, and, when necessary, suspend systems that lead to ethical, legal, or professional violations. At the national level, clubs and federations should adopt existing governance practices from roles such as environmental responsibility and diversity officers by establishing designated personnel responsible for overseeing personal data usage and AI implementation. In this respect, specialized degree programs are already beginning to emerge in leading institutions (<https://www.pace.cam.ac.uk/courses/mst-ai-ethics-and-society> accessed on 8 March 2026), supporting the development of such expertise.

6. Conclusions

Concerns regarding bias, transparency, and accountability in AI have gained substantial attention in recent years across scientific and regulatory domains. It is now widely recognized that data-driven decision-making is not inherently objective and may reproduce or amplify existing inequalities, lack transparency, and raise concerns related to data privacy, accountability, and human oversight. While these risks have been extensively documented in domains such as hiring, education, and healthcare, their implications for sport have only recently begun to be discussed. In this context, the present paper examines ML-driven decision tools in player evaluation and talent identification, highlighting the risks of algorithmic bias, data privacy, and digital determinism, and translating these concerns into operational governance principles tailored to sport settings.

Creating a responsible environment for AI in sport that aligns with principles of good governance and respect for human rights is essential to ensure that technological advances benefit all stakeholders without compromising the integrity of sport [74]. At the same time, the rapid and ongoing evolution of AI calls for a forward-looking approach to governance. As applications expand in ways that are often difficult to anticipate, governance must incorporate foresight, continuous stakeholder engagement, and adaptive integration into decision-making processes [38]. Importantly, AI systems need not inevitably reinforce bias. When deliberately designed and appropriately constrained, they may also mitigate it. For example, Dubois and Walzak [75] demonstrated that a “blind scouting” approach using AI-anonymized match footage shifted evaluative attention toward tactical behavior

and decision-making, reducing reliance on physical appearance and demographic cues. Such findings illustrate that the direction of AI's impact in sport is not predetermined but depends on how these systems are designed, implemented, and governed.

7. Limitations and Future Directions

This paper adopts a policy-oriented perspective grounded in an extensive examination of the literature, based on iterative searches and citation tracking (backward and forward), alongside critical appraisal of source credibility and relevance. However, it does not follow a formal systematic or scoping review protocol and therefore does not aim to provide a fully replicable overview of all available research. In addition, the analysis is constrained by the limited transparency surrounding AI implementation in professional sport, where competitive pressures often lead organizations to restrict disclosure of technologies and practices, thereby obscuring the full scope of current applications. Future research should incorporate qualitative approaches, including interviews, observations, and case studies, to examine how ML-based scores, as an example, are interpreted and integrated into organizational decision-making processes. Such work would clarify how model outputs are used in practice alongside expert judgment and how their influence is conditioned by organizational constraints. Future research should also examine the trade-offs associated with regulating and constraining AI use in highly competitive settings, particularly whether requirements related to transparency, accountability, and fairness impose costs in terms of efficiency, performance optimization, and competitive advantage, and how these tensions are balanced in practice. In addition, prospective longitudinal studies are needed to evaluate the overall performance of ML-based ranking models, including their predictive validity, by tracking both highly ranked and lower ranked prospects within the same developmental systems. Such designs require that athletes across the ranking spectrum receive meaningful opportunities to train, compete, and progress. Retrospective studies could complement this approach by comparing the youth ML-based assessments of current elite and non-elite professional athletes. This would depend on organizations opening historical model records and selection data, for example through collaboration by NCAA programs sharing prior evaluations of high school prospects or NBA clubs providing historical assessments of selected and non-selected draft candidates. Another important avenue would be to examine whether ML-driven talent identification gradually channels development toward a narrower dominant athlete profile, reflected in increasing convergence of anthropometric characteristics over time. Finally, the present analysis primarily extrapolates potential risks associated with ML applications that have been documented in educational, occupational, and medical settings, building on their underlying mechanisms and extending them to professional sport. It remains to be determined whether and to what extent these dynamics will unfold across specific sport disciplines.

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